

Two-Photon Emissive Dyes Based on Push–Pull Purines Derivatives: Toward the Development of New Photoluminescence Bioprobes

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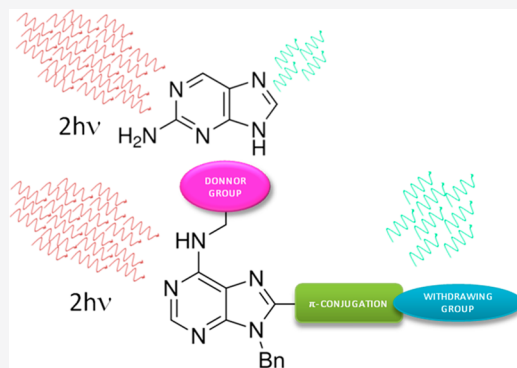


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ABSTRACT: Fluorescent organic molecules have received great attention due to their largest applications, for example, in DNA and RNA spectroscopies studies, development of new photoluminescence bioprobes, and applications in fluorescence spectroscopy. In specific, purine base analog molecules present high fluorescence quantum yields and significant Stokes shift. Furthermore, the addition of push–pull structures at the purine core could increase the photoluminescence properties, making candidates for photoluminescence bioprobes. To consider this, a complete spectroscopic study was performed on nine push–pull purines, distinguished by different push–pull structures. In specific, for this research, the two-photon absorption (2PA) study showed that the compounds present induced two-photon fluorescence at the therapeutic window, desired for fluorescence microscopy. The brightness property was evaluated, indicating that all chromospheres are fluorescent by a 2PA process. Additionally, ultrafast transient absorption was performed to elucidate contribution of the excited states on the 2PA spectra, and quantum chemistry calculations were performed to corroborate the experimental results.



I. INTRODUCTION

Researches related to fluorescence spectroscopy and molecular bioimaging have gained considerable attention over the past decade.^{1–5} Organic molecules with significant fluorescence properties, such as high emission quantum yields (ϕ_f) and large Stoke shifts have been synthesized on a large scale, aiming to improve fluorescence spectroscopy applications.^{6,7} One important application is fluorescence microscopy, in which organic photoluminescence bioprobes can be used as targets to molecular fluorescence imaging in the intracellular environment.^{8–10} At least three important characteristics should be observed in photoluminescence bioprobes: high selectivity, high sensitivity, and low toxicity level. Selectivity is defined by the affinity between recognition elements and receptors,¹¹ while sensitivity is defined by the brightness of the bioprobe emission in the environment that one wishes to carry out the fluorescence spectroscopy.¹¹ In this work, it will be presented a vast photophysical characterization of nine purine molecules, which according to the results, should give ones where we will be able to distinguish the most promising compounds for fluorescence microscopy, for example.

The primary structure of the investigated compounds is an organic backbone that links together a pyrimidine ring and an imidazole ring. Such a backbone includes the nucleobases of nucleic acids (DNA and RNA). Tailored purines can be used

as nucleobases derivatives, and therefore, this class of organic compounds has the potential to be fluorescent nucleobases, allowing spectroscopic studies on DNA and RNA.¹² The main goal of tailoring purines derivatives is to increase the emission quantum yield, as well as to improve other fluorescence properties, such as the induced emission via multiphoton excitation.^{12,13} Several types of purine derivative molecules, which have different peripheral structures, have been studied over the years due to their attractive fluorescence properties^{14–18} in the field of biology, in particular in live cells bioimaging,^{19,20} DNA detection,^{21–23} and fluorescent probes purposes.^{24,25}

DNA or RNA spectroscopic studies are particularly challenging^{26,27} due to their nonradiative decay intrinsic mechanisms, which have a subpicosecond lifetime.^{23,28} To overcome this challenge, compounds with a molecular structure similar to those nucleobases but with emissive features, have been developed.^{12,29} It is linked to the DNA

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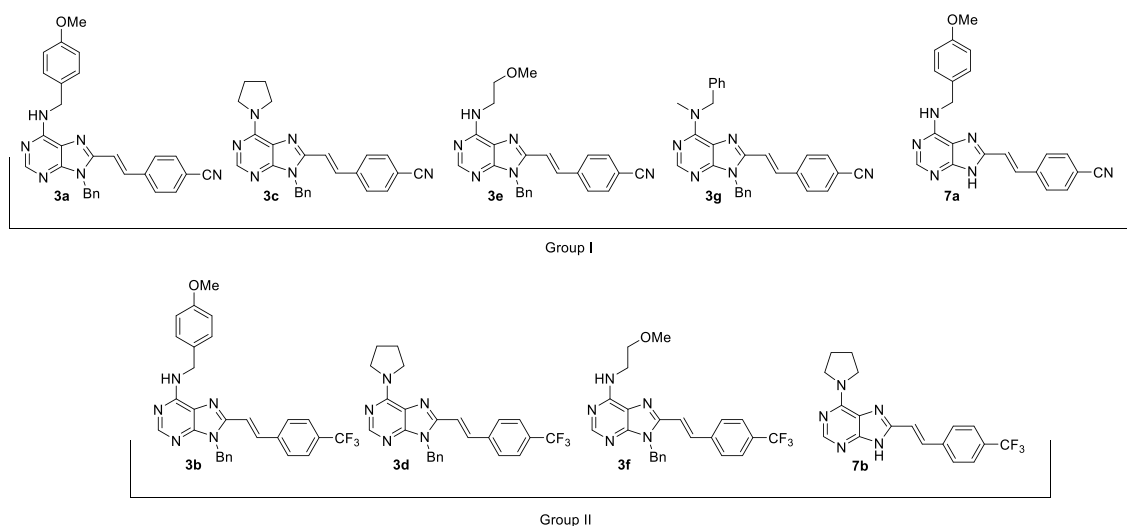


Figure 1. Push–pull 6-amino-8-styryl purines chemical structures.

molecule, resulting in a compound with a radiative decay mechanism but without functionality changes. The 2-amino purine (2AP) is an example of a fluorescent nucleobase, reported as a promising compound to be used on a spectroscopic DNA study.^{23,30,31} The search for similar molecular structures, such as purine derivatives is of extreme importance to the improvement of DNA spectroscopy studies, in particular those structures that possess identical or improved emission properties as the ones reported relative to the 2AP. The emission properties of new compounds could be improved by considering some strategies, such as the increasing of the π -conjugation by adding various push–pull structures at their peripheral molecular structure.³² This strategy could create a larger Stokes shift and increase the fluorescence quantum yield, which will result in the desirable optimization of the fluorescence properties.²⁹

In order to understand the possibility of a new class of purines derivatives, such as the ones reported in this work, to be potential fluorescent nucleobases, linear optical measurements were performed, such as one-photon absorption (1PA), fluorescence emission, fluorescence anisotropy, and fluorescence lifetime, as well as a solvatochromic study. Some of those photophysical parameters combined with the two-photon absorption (2PA) analysis, obtained with a tunable femtosecond Z-scan technique,^{33,34} may reveal the potential of purines derivative as bioprobes. In addition to these studies, new photophysical information can be determined combining the nonlinear optical measurements with the sum over states (SOS) phenomenological model.^{35–38} Femtosecond transient absorptions were performed to identify excited states absorption that could interfere to the 2PA cross-section determination, revealing the excited state absorption regions and the relaxation lifetime of higher excited states that may be involved. Moreover, to understand the some of the determined optical properties, quantum chemical calculations (QCC) were made by using the Gaussian16 software package to simulate the one- and two-photon absorption spectra. These calculations help us to compare the position of the spectral region of interest between the experimental and theoretical results.

II. RESULTS

II.1. Compounds. The full synthesis description of the nine push–pull 6-amino-8-styryl purines studied in this work was reported in the work of Vabre.²⁹ For the sake of consistency, the same nomenclature and classification reported in ref 29 was also used here. Therefore, hence forward the nomenclature of the nine studied samples are **3a** ((*E*)-4-(2-(9-benzyl-6-((4-methoxybenzyl)amino)-9*H*-purin-8-yl)vinyl)benzonitrile), **3b** ((*E*)-9-benzyl-*N*-(4-methoxybenzyl)-8-(4-(trifluoromethyl)styryl)-9*H*-purin-6-amine), **3c** ((*E*)-4-(2-(9-benzyl-6-(pyrrolidin-1-yl)-9*H*-purin-8-yl)vinyl)benzonitrile), **3d** ((*E*)-9-benzyl-6-(pyrrolidin-1-yl)-8-(4-(trifluoromethyl)styryl)-9*H*-purine), **3e** ((*E*)-4-(2-(9-benzyl-6-((2-methoxyethyl)amino)-9*H*-purin-8-yl)vinyl)benzonitrile), **3f** ((*E*)-9-benzyl-*N*-(2-methoxyethyl)-8-(4-(trifluoromethyl)styryl)-9*H*-purin-6-amine), **3g** ((*E*)-4-(2-(7-benzyl-4-(benzyl(methyl)amino)-7*H*-pyrrolo[2,3-*d*]pyrimidin-6-yl)vinyl)benzonitrile), **7a** ((*E*)-4-(2-(6-((4-methoxybenzyl)amino)-9*H*-purin-8-yl)vinyl)benzonitrile), and **7b** and (*E*)-6-(pyrrolidin-1-yl)-8-(4-(trifluoromethyl)styryl)-9*H*-purine).

All the molecular structures of the studied samples are depicted in Figure 1. The structures show an amine function (N) on position 6 of the purine scaffold substituted by four different groups, which are a pyrrolidine, a methoxy (OMe) ethyl chain, *p*-methoxyphenyl ring, and a phenyl ring. Overall, these four amino substituents have an electron-donor character. At position 8 of the purine, a styryl group will act as the π -conjugated linker to one of the electron-withdrawing groups, which could be a cyano (CN) or a trifluoromethyl (CF₃) group at the *para* position of the styryl group. In summary, the studied compounds will always have one electron donor group, a purine structure, and an electron-withdrawing group featuring a push–pull arrangement. The major distinction between each compound is the difference between the strength of the electron-withdrawing group linked to the *para* position of the styryl group and the donor groups that are linked at the amine group. As can be seen in Figure 1, the molecular structures of compounds **7a** and **7b** do not present a protecting group,^{39,40} the benzyl (Bn) group, on position 9 of the purine structure.

The investigated compounds were classified into two groups: Group I comprises all compounds bearing the CN in

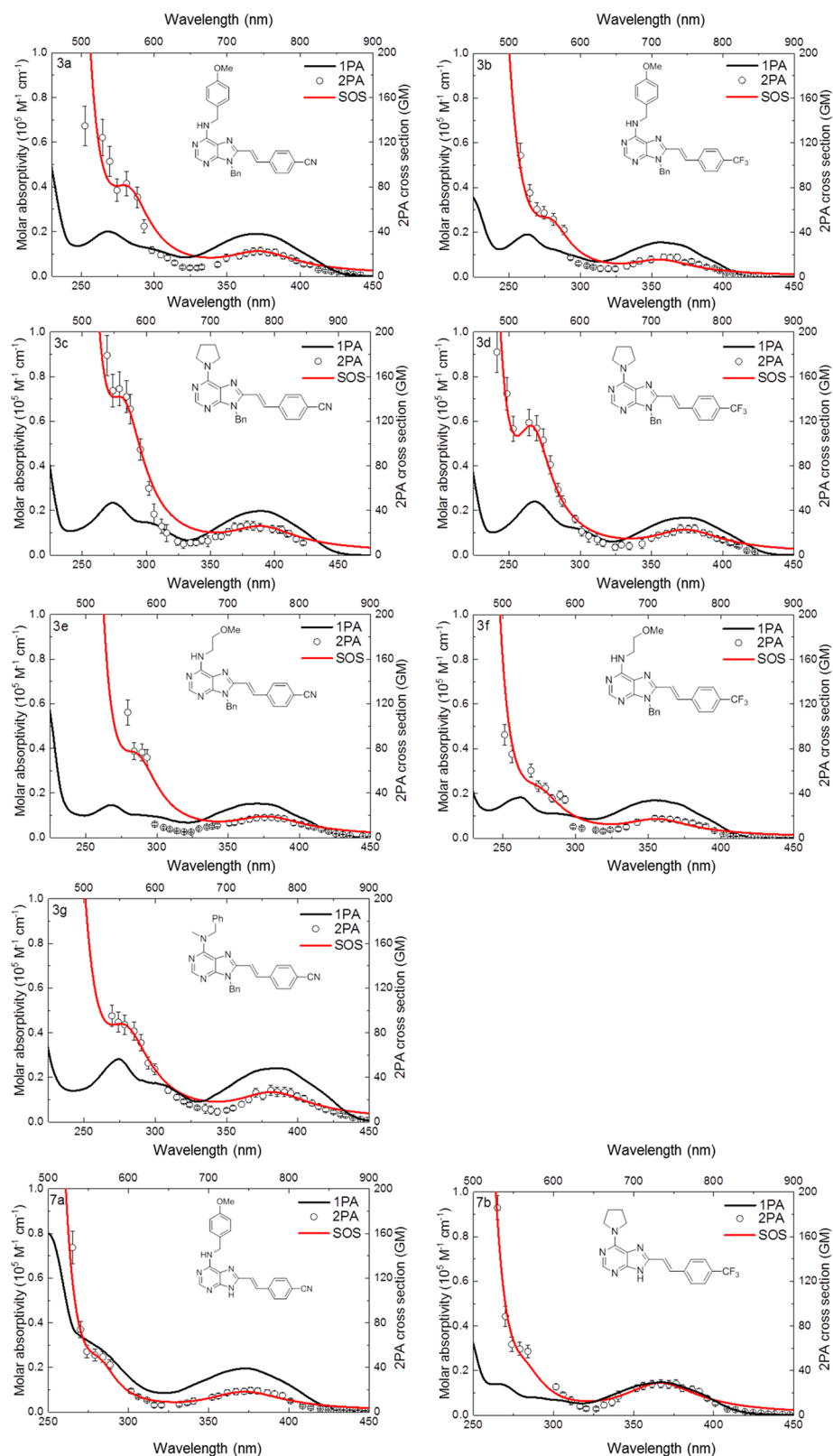


Figure 2. 2PA cross sections (black circles), SOS model fittings (solid red lines) and molar absorptivities (solid black lines) of the nine purines dissolved in DCM (3a–3g) and DMSO (7a and 7b).

the *para* position of the styryl group, such as 3a, 3c, 3e, 3g, and 7a; group II includes the compounds 3b, 3d, 3f, and 7b, which have the CF₃ in the *para* position of the styryl group.

II.2. Linear and Nonlinear Optical Measurements. All compounds in its powder form were dissolved by using

spectroscopic grade solvents. The molecular structures from 3a to 3g were dissolved in dichloromethane (DCM), while 7a and 7b, were dissolved in dimethyl sulfoxide (DMSO), and kept in fused quartz cuvettes. Both molar absorptivity and fluorescence emission spectra of all samples are presented in the Supporting

Information (Figure S1), while the two-photon absorption spectra are presented in Figure 2. The remaining experimental methods, procedures and figures, such as, fluorescence lifetimes, fluorescence anisotropy, solvatochromism, Z-scan curves, and transition absorption are presented in the Supporting Information.

Three electronic transitions in the spectral region from 250 to 450 nm (from 4.96 to 2.76 eV) are common for all the compounds, as could be observed in Figure S1. The intensities of those electronic transitions are represented by molar absorptivities of the order of $10^4 \text{ L cm}^{-1} \text{ mol}^{-1}$, which are in good accordance with previous studies.^{14,41} Some compounds present bathochromic shifts up to 31 nm between their absorbance bands, caused by the different types of push–pull structures added to the purine core. In most cases, the stronger the electron-donor and -withdrawing groups are, the higher the bathochromic shift is. Like mentioned before, all the studied compounds have a push–pull arrangement donor- π -withdrawing, thus distinct absorptions bands are expected according to the different substituents playing the role of electron-donor or -withdrawing. Due to that fact, one could observe that the bathochromic shift is higher in the compounds bearing a strong electron-withdrawing group (CN) on the *para* position of the phenyl ring when compared with the compounds from group II, i.e., with CF_3 group at the same position.

The fluorescence spectra were obtained by exciting the samples at 350 nm. The spectral position of the maximum emission, for all samples, varies between *ca.* 430 and 500 nm (2.88 and 2.48 eV), according to the polarity of the solvent used, as shown in Figure S1 (solid red lines). By comparing the maxima intensity between absorption and emission bands, one could notice a spectral difference of approximately 100 nm for every compound, revealing a large Stokes-shift. The magnitude of such spectral difference suggests a significant energy difference between the Franck–Condon excited states probably due to vibrational, electronic or geometric differences.²⁹ Table S1 in the Supporting Information presents the molar absorptivity and wavelengths corresponding to absorption band peaks, as well as wavelengths corresponding to fluorescence emission peaks.

The fluorescence anisotropy (r)^{42,43} measurements were performed by acquiring the fluorescence emission at 480 nm, which is approximately the emission peak of all studied samples. The solvent used to determine the fluorescence anisotropy was the same used for the absorbance measurements, i.e., DCM (3a to 3g) and DMSO (7a and 7b) and the results are shown in the Figure S3. It is important to highlight that for “pure” electronic transitions, in an ideal case, the excitation anisotropy remains almost constant. Inhomogeneous broadening and overlapping between excited states are aspects that may be considered for changes in the values of anisotropy. In our particular case, the excitation anisotropy values are almost constant from 350 to 400 nm (3.54 to 3.10 eV), suggesting that such an absorption band corresponds to a transition to the first excited state $S_0 \rightarrow S_1$.

Six solvents were used in the solvatochromism measurements: acetone, ethanol, methanol, DCM, DMSO, and toluene. As expected, the lower absorption energy band (centered *ca.* at 350 nm) was shifted according to the solvent polarity.^{44,45} The same spectral shift behavior was identified in the fluorescence emission bands; i.e., there is a Stokes shift according to the solvent and the different strengths of the

molecule's dipole moments. The Stokes shifts of all studied samples are illustrated in Figure S4. Once the excited state lifetime ($\tau_f \sim 3 \text{ ns}$) is longer than the charge rearrangement of the system, both molecule and solvent have enough time to perform the charge rearrangement aiming for a decrease in the energy of the system, which results in the fluorescence emission shifts, according to solvent polarity. The difference between the maxima absorption and emission wavelengths ($\Delta\nu$), corresponding to the compounds 7a and 7b, here represented in cm^{-1} were obtained from the Vabre work.²⁹ Through linear fittings, depicted in Figure S5, angular coefficients ($\frac{\Delta\nu}{\Delta F}$ in cm^{-1}) were determined for all samples and are shown in Table 1.

Fluorescence lifetimes (τ_f) were determined, and the values are shown in Table 1. The experimental lifetime decay curves and their respective monoexponential fittings are shown in the Supporting Information (Figure S2). The obtained τ_f values ranging from 3 to 4 ns are in accordance with the ones reported by Butler and co-workers¹⁴ for other types of purine molecules, such as the donor–acceptor 2-aminopurine (2AP) based-molecules. The 2AP molecule is a purine derivative fluorescent analog used as a substituent of adenine for DNA properties studies and has a monoexponential lifetime decay of about 10 ns.³¹ However, push–pull 6-amino-8-styryl purine molecules' fluorescence lifetimes are shorter than that for the 2AP molecule, indicating that the insertion of both π -conjugated links and the push–pull groups into the purine core affect the relaxation lifetime. Different relaxation lifetimes can be explained by Fermi's golden rule.^{46,47}

For an emissive compound to be considered a promising bioprobe, it is fundamental to have a high fluorescence quantum yield (ϕ_f). All studied compounds have ϕ_f values above 60% reported by Vabre.²⁹ The reported 2AP ϕ_f values are about 68%,²³ revealing a good agreement with the ones reported and confirmed in this work, which have the values ranging from 62% to 99%. Usually, emissive compounds with high ϕ_f have an effective radiative decay from the first excited state to the ground state.^{43,48} In other words, when the value of the radiative decay (k_r) is much higher than the value of the nonradiative decay (k_{nr}), it means that the decay is more likely to occur by a photon emission than by vibrational relaxation. Those values can be estimated by using the fluorescence lifetime and quantum yield values experimentally obtained.⁴⁹ The estimated values of k_r and k_{nr} are presented in the Table 1, and the calculation details are enunciated in the Supporting Information. The highest difference between k_r and k_{nr} is associate to group II, in particular, to the compound 7b, as excepted because of its high ϕ_f .

Although, for a bioprobe to be triggered by two-photon excitation, the high value of ϕ_f alone could not be enough. In this particular case, another photophysical parameter must be considered—brightness.^{50,51} To estimate the brightness of the studied compounds, one must acquire the 2PA cross-section at the desirable excitation wavelengths and multiply it for its fluorescence quantum yield.

The permanent electric dipole moment ($\Delta\mu_{01}$) was estimated by applying the Lippert–Mataga equation.^{42,43} All the physical parameters needed to use such an equation were acquired and calculated with the linear optical measurements described so far, except one, the molecular cubic radius (a^3). To determine a^3 , we have assumed that each molecular structure occupies a spherical volume ($\text{Vol}_i = \frac{4}{3}\pi a^3$). Applying

Table 1. Spectroscopic Parameters Obtained by Linear Optical Measurements of All Investigated Compounds^a

	τ_f (ns)	ϕ_f (%)	k_r (ns ⁻¹)	k_{ns} (ns ⁻¹)	r	$\Delta\nu/\Delta F$ (cm ⁻¹)	a^3 (Å ³)	$\Delta\mu_{01}$ (D)	μ_{01} (D)	μ_{02} (D)	μ_{03} (D)
3a	2.9 ± 0.3	75	0.26 ± 0.02	0.08 ± 0.01	0.015 ± 0.002	2124 ± 749	250 ± 50	10.2 ± 2.5	5.3 ± 0.6	4.3 ± 0.5	4.9 ± 0.5
3b	3.0 ± 0.3	81	0.27 ± 0.02	0.06 ± 0.01	0.015 ± 0.002	1872 ± 497	270 ± 60	9.9 ± 2.0	4.6 ± 0.5	3.6 ± 0.4	4.4 ± 0.5
3c	3.6 ± 0.4	62	0.17 ± 0.01	0.11 ± 0.01	0.013 ± 0.002	2704 ± 547	280 ± 90	12.4 ± 2.8	5.5 ± 0.6	3.4 ± 0.4	6.0 ± 0.6
3d	3.7 ± 0.4	79	0.21 ± 0.02	0.06 ± 0.01	0.010 ± 0.002	2070 ± 645	220 ± 40	9.7 ± 2.2	5.0 ± 0.5	2.7 ± 0.3	6.0 ± 0.6
3e	3.0 ± 0.3	63	0.20 ± 0.02	0.13 ± 0.01	0.014 ± 0.002	3187 ± 686	250 ± 60	12.5 ± 4.0	4.7 ± 0.5	3.6 ± 0.4	4.0 ± 0.4
3f	3.0 ± 0.3	65	0.21 ± 0.02	0.12 ± 0.01	0.018 ± 0.003	1820 ± 261	320 ± 100	10.7 ± 3.5	5.0 ± 0.5	3.4 ± 0.4	5.0 ± 0.5
3g	3.6 ± 0.4	63	0.18 ± 0.01	0.10 ± 0.01	0.010 ± 0.002	876 ± 224	220 ± 60	6.10 ± 2.1	6.0 ± 0.6	3.7 ± 0.4	6.7 ± 0.7
7a	2.9 ± 0.3	81	0.28 ± 0.03	0.064 ± 0.006	0.044 ± 0.008	5669 ± 2222	180 ± 50	14.1 ± 4.2	5.2 ± 0.5	4.9 ± 0.5	4.0 ± 0.4
7b	3.1 ± 0.3	99	0.32 ± 0.03	0.0026 ± 0.0002	0.043 ± 0.007	3532 ± 846	190 ± 60	11.4 ± 3.5	4.8 ± 0.5	3.0 ± 0.3	3.2 ± 0.4

^aAll the values of ϕ_f and the values $\frac{\Delta\nu}{\Delta F}$ of compounds 7a and 7b were obtained from ref 29.

the Smoluchowski–Einstein diffusion equation,^{52,53} the values of a^3 , in Angstrom units, were obtained, and they are given in Table 1. The estimated a^3 is around 250–300 Å³, for all molecular structures. Moreover, the cubic radius was also calculated by using the Gaussian 16 program package,⁵⁴ which revealed good accordance with the experimentally derived values, taking into account the experimental error, as shown in Table S21. With all the determined values of a^3 , r , τ_f and $\frac{\Delta\nu}{\Delta F}$, it was possible to estimate the $\Delta\mu_{01}$ range values, presented in Table 1 and expressed in Debye units.

Lorentzian decompositions of the absorption coefficient α_0 (ν) spectra, as shown in Figure S16, were made to estimate the transition dipole moments μ_{01} , μ_{02} and μ_{03} . The values of μ_{01} , μ_{02} , and μ_{03} do not show a considerable dispersion as one could check in last three rows of Table 1. However, it can be noted that in the samples of group I there is a slight tendency of μ_{01} , μ_{02} , μ_{03} , and $\Delta\mu_{01}$ be higher than samples of Group II. This tendency was expected and is in agreement with the 1PA bands. The samples with a CN as an electron-withdrawing group present 1PA bands located in longer wavelengths (lower energy) and are those with higher electric dipole moments, thus, present the lowest energy between the excited and the ground states. On the contrary, when the electron-withdrawing group is CF₃, it is possible to notice that, due to the dipole moments tending to lower values, the 1PA bands are located in regions of shorter wavelengths (higher energy), in this way, as already mentioned, the differences between excited and ground states are higher than when the electron-withdrawing group is CN.

Once obtained the linear optical parameters, the nonlinear optical measurements started with the acquisition of the 2PA cross-section as a function of the wavelength ($\sigma_{2PA}(\lambda)$). It is expected that the spectral shape of the ($\sigma_{2PA}(\lambda)$) follows the 1PA analog due to the lack of charge distribution symmetry of the molecular structure, i.e., the same energy levels excited by 1PA should also be excited by 2PA, as described by the selection rules.^{55,56} The black circles in Figure 2 represent the acquired 2PA, which revealed that the absorption induced by two-photons also permits the same energy transitions allowed by one-photon. The ($\sigma_{2PA}(\lambda)$) values of interest are in the spectral region of 650–850 nm (1.91–1.46 eV), in particular, due to the highest penetration of this range of wavelengths when compared with shorter ones.^{57,58} At the resonance enhancement region, i.e., for wavelengths shorter than 600 nm (2.07 eV), $\sigma_{2PA}(\lambda)$ values are higher; however, it could not bring such an advantage to bioimaging applications.

The 1PA and 2PA joint analysis make it possible to fit the ($\sigma_{2PA}(\lambda)$) based on the sum-overstates (SOS) approach,^{35–37} leading to the evaluation of more spectroscopic parameters, such as permanent electric dipole moments related to other electronic transitions than the $\Delta\mu_{01}$. This analysis is carried out considering only the electronic transitions between the energy levels. Some 2PA studies include vibronic transitions, in particular in complex protein environments,⁵⁹ which is not the present case. The experimental ($\sigma_{2PA}(\lambda)$) spectra were fitted employing the following SOS equation:

$$\sigma_{2PA}(\lambda) = \frac{2}{5} \frac{(2\pi)^4}{(chn)^2} L^4 \frac{1}{\pi} \left[\frac{(\Delta\mu_{01}^2 \cdot \mu_{01}^2) \cdot G_{01}}{(\nu_{01} - 2\nu)^2 + G_{01}^2} + \frac{(\Delta\mu_{02}^2 \cdot \mu_{02}^2) \cdot G_{02}}{(\nu_{02} - 2\nu)^2 + G_{02}^2} + \frac{(\Delta\mu_{03}^2 \cdot \mu_{03}^2) \cdot G_{03}}{(\nu_{03} - 2\nu)^2 + G_{03}^2} + \frac{\nu^2}{(\nu_{01} - \nu)^2 + G_{01}^2} \frac{(\mu_{13}^2 \cdot \mu_{01}^2) \cdot G_{03}}{(\nu_{03} - 2\nu)^2 + G_{03}^2} + \frac{\nu^2}{(\nu_{01} - \nu)^2 + G_{01}^2} \frac{(\mu_{14}^2 \cdot \mu_{01}^2) \cdot G_{04}}{(\nu_{04} - 2\nu)^2 + G_{04}^2} \right]$$

in which, $L = \frac{3\pi^2}{2n^2 + 1}$ is the Onsager factor, G_{nm} is the half-width at half-maximum of the Lorentzian deconvolutions that, in this particular case, has the value of about 0.24 eV for all transitions (see Figure S6). $\Delta\mu_{nm}$ is the difference between excited states ($m = 1, 2$, or 3) and ground state ($n = 0$) permanent electric dipole moment, μ_{ki} is the transition electric dipole moments, and ν_{nm} are the transitions frequencies. The values of μ_{01} , μ_{02} , μ_{03} , and $\Delta\mu_{01}$, experimentally determined by 1PA and solvatochromism, were used as input parameters in the SOS model to evaluate $\Delta\mu_{02}$, $\Delta\mu_{03}$, μ_{13} , and μ_{14} , displayed in Table 2. The SOS fitting results are represented by solid red

Table 2. $\Delta\mu_{02}$, $\Delta\mu_{03}$, μ_{13} , and μ_{14} Determined Through SOS Fitting on the 2PA Cross Section Spectra ($\sigma_{2PA}(\lambda)$)^a

	3a	3b	3c	3d	3e	3f	3g	7a	7b
$\Delta\mu_{02}$ (D)	3.2	3.2	3	3.3	3.0	3.5	2.0	3.1	3.2
$\Delta\mu_{03}$ (D)	4.0	4.0	5.6	5.8	4.1	4.2	2.5	3.3	4.3
μ_{13} (D)	5.2	3.7	5.6	4.5	4.3	4.8	4.2	4.4	4.2
μ_{14} (D)	11	14	14	12	14.2	11	10	9.5	11

^aThe error associated to the quantities obtained by the SOS approach are around 20%. D represents debye units.

lines in Figure 2. It is important to highlight here that $\Delta\mu_{02}$ and $\Delta\mu_{03}$ were also determined by using solvatochromism measurements (all the calculation details can be found in the Supporting Information). The values shown in Table 2, considering the experimental and fitting errors, are in reasonable agreement with the ones depicted in Table S2.

The transition dipole moments and the difference between permanent dipole moments obtained with the SOS model adjusting the 2PA experimental spectra demonstrated a low dispersion of the values when compounds are compared from each other. It indicates that the compounds have the similar dipolar character. Also, it is observed that these low dispersions are similar to the ones depicted in Table 1, with respect to μ_{01} , μ_{02} , μ_{03} , and $\Delta\mu_{01}$. In specific, $\Delta\mu_{02}$ and $\Delta\mu_{03}$ are in agreement, considering the experimental and fit errors, to the ones displayed in Table S2.

Moreover, transient absorption measurement (see Supporting Information) was used to determine the excited state absorption spectra. Its main proposal is to check for the presence of excited states. Therefore, through this technique, it is possible to explain if the 2PA may have contribution from excited state absorptions. The transient absorption measurements have shown the presence of an excited state in all molecules, see Figure S8, in a spectral region that covers 540–640 nm. Thus, comparing this result with the 2PA spectrum,

one can say that in such a spectral region (540–640 nm) 2PA spectra may have contribution from excited state absorption. On the other hand, for the 2PA lower energy band (700–850 nm), there is not excited state absorption, which indicates a pure 2PA band. In addition, transient absorption allowed us to verify the excited state lifetimes from the temporal pulse duration to 1 ns (corresponding to the maximum size of the translational stage of pump and probe technique; see Supporting Information). In this way, it was possible to observe that the higher excited state shows to have a lifetime in the order of the temporal pulse duration. The plateau, in the excited state absorption, describes a longer lifetime for the first excited state, a fact that corroborates the measurements obtained via time-resolved fluorescence technique.

A previous work,⁵ inferred the $\sigma_{2PA}(\lambda)$ values of the compound 2AP in four solvents (water, methanol, DMSO and glycerol/water) by using a TPE fluorescence technique. The maximum reported $\sigma_{2PA}(\lambda)$ value associated with compound 2AP is 0.20 GM, in the spectral window from ca. 600 to 650 nm (2.07 to 1.91 eV). One can compare the cross-section values induced by two-photons between 2AP and the purine derivatives presented in this work and will notice that the latter ones possess a cross-section 100 times higher than the former one. Such a difference of magnitude could be explained by the structural differences between 2AP and the purine derivatives. The molecular structure of the 2AP is a combination between a pyrimidine and an imidazole ring, which has a shorter conjugation length when compared to the purine derivatives. The addition of the push–pull and the styryl groups increased the conjugation length of the molecules, giving them a higher dipolar character, which could trigger an increase in the $\sigma_{2PA}(\lambda)$ values. Another interesting fact is the redshift associated with the purine derivatives, while compared with the 2AP compound. While the 2AP bands are centered approximately between 600 and 650 nm (2.07–1.91 eV), the purine derivatives bands are centered at 700–800 nm (1.77–1.55 eV), which is an attractive spectral therapeutic window.

To verify if the emission of the investigated compounds could be purely triggered by two-photon excitation (2PE), fluorescence measurements via 2PA were performed. The experimental apparatus used is similar to the one used in the time-resolved fluorescence technique. The primary difference resides in the method of fluorescence spectrum acquisition, which in this case, was obtained by an optical fiber that carries the signal to a spectrophotometer. This procedure allows discriminating the fluorescence from the excitation peak intensities. All samples were excited at 780 nm with a 150 fs pulse-width at different intensities. The fluorescence emission as a function of the excitation pulse energy was acquired and is depicted in Figure 3, showing a quadratic behavior (slope = 2), proving that a purely 2PE process can be triggered for both groups of the investigated compounds.

As mentioned before, the product between the fluorescence quantum yield and the two-photon cross-section at a specific wavelength $\phi_f \times \sigma_{2PA}(\lambda)$ retrieves the value of a spectroscopic parameter called brightness.^{50,51} The selected wavelengths to evaluate the brightness were 560 and 780 nm (2.21 and 1.59 eV), as depicted in Figure 4. The shorter wavelength excitation (560 nm, 2.21 eV) presented higher brightness values while compared with the longer wavelength (780 nm, 1.59 eV). However, the longer wavelength excitation is more favorable for bioimaging applications.^{57,58} The highest brightness value

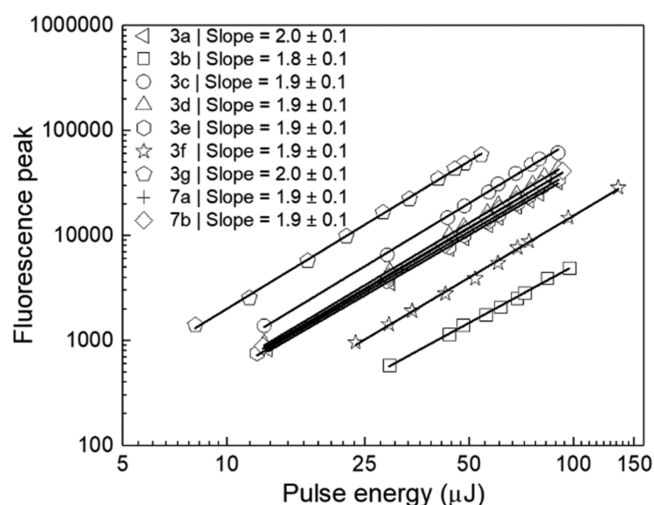


Figure 3. Induced fluorescence by 2PA (780 nm), fluorescence peak versus pulse energy, to determine the slope for all samples.

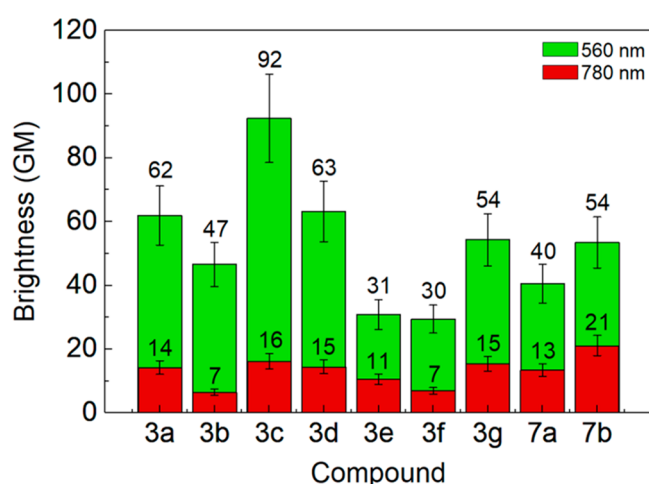


Figure 4. Brightness values for all investigated compounds determined at 560 (green) and 780 nm (red).

in group I is shared by three compounds (3a, 3b, and 3c), while in group II, the highest value belongs to compound 7b.

Finally, using the values of 2PA cross-section for 2AP dissolved in water,⁵ and the value of 2AP fluorescence quantum yield (also dissolved in water),²³ we are able to estimate the 2AP brightness. This value corresponds to 0.136 GM, 47–150 times lower than our molecules. It is important to say that the value of brightness, for 2AP and our samples, corresponds to the maximum 2PA of the lower energy band. The addition of the styryl and push–pull groups increased the conjugation length and the dipolar character of the molecules, which could trigger an increase in brightness values.

II.3. Simulation of the One- and Two-Photon Absorption. A postprocessing software was used to achieve the simulate spectra, both one- and two-photon absorption spectra, based on parameters calculated by using Gaussian 16 package (Rev. B.01).⁵⁴ The entire methodology regarding the theoretical calculations used in this work is described here.⁶⁰ The quantum-chemical calculations which allowed to obtain such parameters were carried out at the TD-DFT theory level. First, the equilibrium geometry of every single molecule was calculated to obtain their optimized geometry. The molecular

structures were optimized at the CAM-B3LYP/6-311+G(d,p) level of theory, in both gas-phase and solvent medium employing a polarizable continuum model (IEF-PCM),⁶¹ using the integral equation formalism variant (IEF-PCM).⁶¹ All optimized geometries are presented in Tables S3–S11. Second, time-dependent calculations were performed by using the optimized geometry, with the same functional and basis set used in the previous procedure, in gas-phase and solvent medium. Tamm–Dancoff approximation (TDA)⁶² was used to calculate the transition dipole moments from the ground state to the j -th excited state (μ_{0j} , $j > 0$), transition dipole moments between excited states (μ_{ij} , $j > 0$ and $j \neq i$), permanent dipole moments in the ground state (μ_{00}) and excited states (μ_{jj} , $j > 0$) to obtain the dipole moments difference ($\Delta\mu_{0j} = \mu_{jj} - \mu_{00}$, $j > 0$), and the excitation energies. By the last point, the simulated spectra were computed by using the lowest 15 excited states, and the SOS formulation used for the simulation of the spectral shape was reported previously.⁶⁰ The 15 excited states used are presented in Tables S12–S20, while the simulated spectra, in both gas-phase and solvent mediums, are depicted in Figure S9.

The experimental and theoretical molar absorptivities in the solvent medium, shown in Figure S10, were plotted as a function of energy to visualize better the energies differences between the simulated and the experimental data. Moreover, the depicted simulated spectra were plotted in its original and shifted version, in which the shifted spectra (solid blue lines) were translated to match the maxima intensity of the experimental lower energy band of every compound (solid black lines). As one could check, the minimum and maximum bathochromic shifts have the values of 0.15 and 0.25 eV, respectively. Such shifting magnitude reveals that the level of theory used (CAM-B3LYP/6-311+G(d,p)) to calculate the 1PA spectra have overestimated the calculated energy levels of the allowed electronic transitions. For example, compound 3a has presented a calculated HOMO–LUMO electronic transition centered at *ca.* 3.54 eV (350 nm), while the experimental absorption band is centered at *ca.* 3.36 eV (370 nm). On the other hand, one of the highest energy overestimations belongs to compound 3g, in which the calculated HOMO–LUMO electronic transition is centered at *ca.* 3.53 eV (351 nm), while the experimental absorption band is centered at *ca.* 3.27 eV (380 nm).

The same procedure was used for the 2PA spectra; i.e., the simulated and experimental two-photon cross-section was plotted as a function of energy, as well as the shifted simulated spectra with the same shift value, as shown in Figure 5. An energy overestimation from the calculated to the experimental 2PA was also found, as expected, once molecular structures with low centrosymmetric charge distribution usually have their two-photon electronic transitions following their one-photon analog.^{63–65}

The primary purpose of the QCC, in this particular study, was to confirm the existence of a leading electronic transition in the spectral region of interest from 650 to 850 nm (from 1.91 to 1.46 eV) to trigger the absorption and excitation via two photons. In this region of interest, the most considerable difference between the theoretical and the experimental 2PA cross sections are around two to three times, which is a significant difference. Such discrepancy between cross sections obtained through QCC simulations is a well-known and widely discussed topic that can be found elsewhere.^{66–68} However, we would like to emphasize that due to the use of ultrashort two-

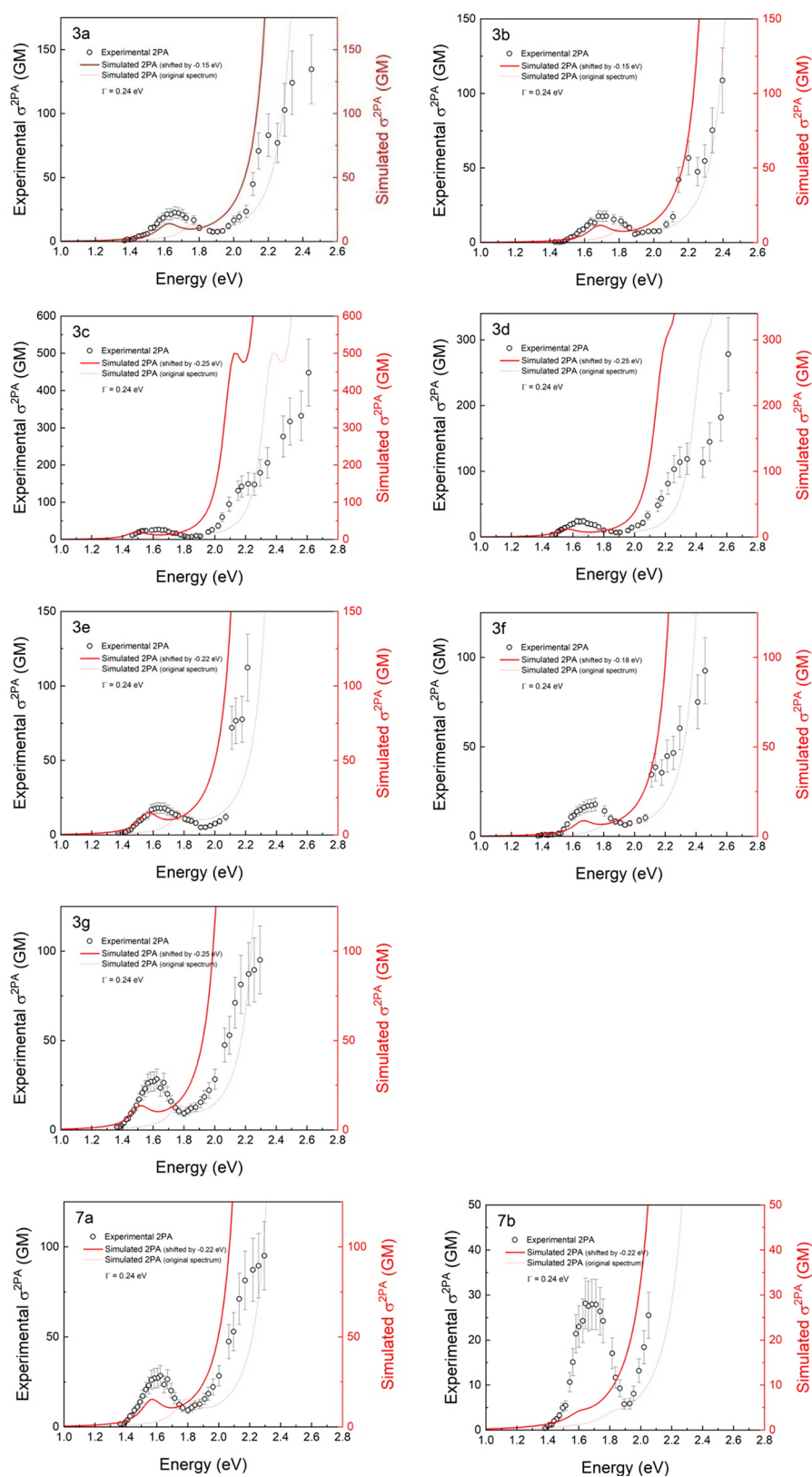


Figure 5. Two-photon absorption simulated and experimental spectra of all studied samples in solvent media. Simulated (red solid and dashed lines, right axis) and experimental (open circles, left axis) 2PA. The solid red lines represent the simulated 2PA spectra plotted with a bathochromic shift between 0.15 and 0.25 eV. The red dashed solid lines are the original simulated 2PA spectra. The value of the damping constant (Γ) used in all simulations corresponds to 0.24 eV.

photon excitation pulses (femtosecond pulses) and low-optical average power ($<100 \mu\text{W}$) when pumping the samples, we are

confident that the most reliable cross sections values are the ones corresponding to the experimental measurements.

Overall, such results suggest that the level of theory have a reasonable agreement with experimental data, in terms of spectral position.

III. CONCLUSIONS

Linear and nonlinear optical studies were performed, revealing the potential of a set of push–pull purines derivatives as a new generation of biological probes via 2PE. The addition of distinct push–pull and styryl groups linked to the purine core increased the 2PA cross-section values comparing to the 2AP molecule previously reported. Furthermore, one of the main results of this work shows that the 2PA spectra are red-shifted as compared with the 2AP, which is a significant advantage for therapeutic applications. This shift was obtained by increasing the conjugation length combined with the strong dipolar character of the push–pull groups on the investigated compounds. One of the main 2PA absorption bands has its maximum cross-section center around 750 nm, allowing better penetration of the laser beam, for example, in mammal tissue. Such a fact could increase the signal–noise ratio when acquiring the fluorescence emission *in vivo* studies, by using the channels of a charged coupled device. Another crucial parameter evaluated in work was the brightness property, in two distinct wavelengths, showing the possibility to tune the excitation to improve the fluorescence signal triggered by 2PA. Quantum chemical calculations confirm the 2PA allowed electronic transition at the spectral region of 650 to 850 nm (1.91 to 1.46 eV), responsible for triggering the excitation via two photons.

In summary, this study showed that these purine derivatives could be promising photoluminescence bioprobes via 2PA. Further spectroscopic studies, for example, two-photon microscopy in DNA and RNA samples, should be performed to confirm the implementation of these types of compounds in bioimaging applications.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcc.0c01859>.

Experimental section, including the experimental description and results as well as the graphs of molar absorptivity and fluorescence emission spectrum, fluorescence lifetimes, solvatochromism measurements, transition electric dipole moment, two-photon absorption (2PA) cross section (obtained through Z-scan technique) and transient absorption (obtained through femtosecond pump and probe technique); description of nonlinear spectroscopic techniques used for determine the 2PA cross-section and the transient absorption; and computational details, which include the simulation of the one- and two-photon absorption spectra in solvent and gas-phase and optimized geometries, excited states, transition energy, and oscillator strength (PDF)

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Notes

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